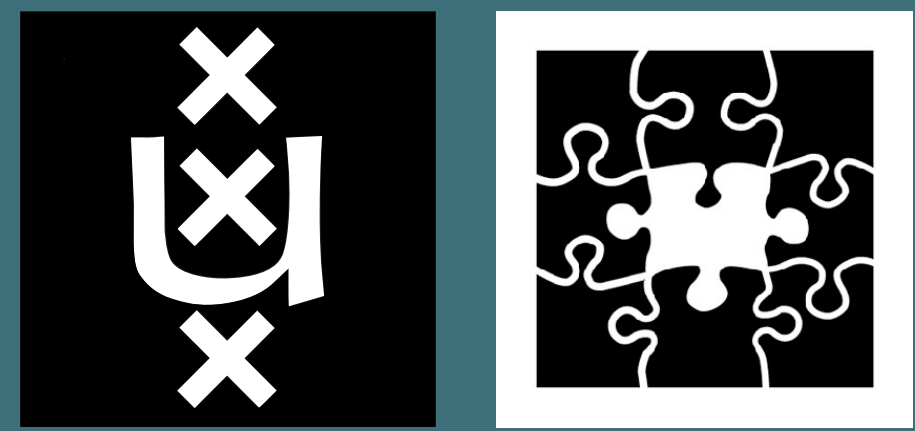




Who is the richest club in the championship?

Detecting and Rewriting Underspecified Questions Improve QA Performance

Motivation

LLMs perform well on clearly-specified queries, yet QA benchmarks remain far from solved. We argue a key culprit is **underspecification (UND)**: questions whose intended meaning *cannot be uniquely determined* without additional context.

Original: “Who is the richest club in the championship?”
Which championship? Richest by what metric? At what time?

Rewritten: “Based on the most recent financial reports, which club in the English Premier League generated the highest revenue?”

Without rewriting: **GPT-4o answers “Leicester City” ×**

After rewriting: **GPT-4o answers “Manchester City” ✓**

4-Step Pipeline

Step 1 — UND Classifier: Select best LLM-classifier from 10 candidates. **Qwen3-4B** wins on both UNDER (855 Q) and expert-verified UNDER-GOLD (150 Q). Avg: acc. 0.74, macro-F1 0.73.

Step 2 — Assess QA on FS/UND subsets: Apply classifier to QA-ensemble (NQ, HotpotQA, TriviaQA, FRAMES; 3,824 Q). Run GPT-4o and Gemini-2.5-Flash on each partition — **questions only, no context passages provided.**

Step 3 — Rewrite UND questions: Oracle rewriter (gold answer + classifier reasoning) converts UND → FS while holding answers fixed. 64–86% of rewrites classified as FS.

Step 4 — Reassess QA: Cross-assignment setup (GPT-4o rewrites → Gemini answers, vice versa). Compare SotA LLM performance scores on rewritten vs. original UND.

Taxonomy of Underspecification Types

Type	Example	Source
1. Missing components	“What’s the capital?” <i>Of which country?</i>	CoCoNot
2. Undetermined reference	“When did the Giants reach the playoffs?” <i>Football or baseball Giants?</i>	CLAMBER
3. Undetermined granularity	“When did WWI break out?” <i>Politically or militarily?</i>	AmbigNQ
4. Undetermined standard	“Best smartwatches in 2023?” <i>Best by what standard?</i>	IN3

Conclusion, Limitation and Repository

- ▶ UND is a **widespread but overlooked** confound affecting the reliability of QA evaluation.
- ▶ LLMs consistently struggle with UND questions with no context available, but performance significantly improves once the UND components are resolved -> **many apparent QA failures reflect question underspecification rather than model limitations.**

The Prevalence of Underspecification

- ▶ UND questions are present in **all four** benchmarks tested:

Dataset	% UND	Sample size
TriviaQA	15.9%	1,000
HotpotQA	49.6%	1,000
NQ	45.8%	1,000
FRAMES	53.4%	824

- ▶ LLMs perform **significantly worse** on UND vs. FS subsets across all datasets and models (independent *t*-tests, all $p < 0.05$; trends hold under both token-F1 and Nvidia AA, an LLM-as-judge metric.).

Rewriting Improves LLM QA Performance on UND questions (F1 Scores shown here)

	NQ	HotpotQA	TriviaQA	FRAMES
GPT-4o orig.	37.0%	34.6%	75.8%	24.4%
GPT-4o rewr.	57.3% (+20.3)	51.8% (+17.2)	83.6% (+7.8)	41.6% (+17.2)
Gemini orig.	38.8%	41.2%	76.0%	37.1%
Gemini rewr.	50.0% (+11.2)	50.6% (+9.4)	74.4% (−1.6)	46.5% (+9.4)

- ▶ TriviaQA exception expected: Trivia quiz questions relatively more self-contained by design. Rewr. AA still increases for both GPT-4o and Gemini-2.5-Flash.
- ▶ Lexical overlap as a sanity check rules out answer leakage from rewriting (Jaccard < 0.1 , n-gram < 0.11 across all configs).

Rewriting in Action

Q1 Original: “Who owns the majority of the US debt?”

Gemini-2.5-Flash: “The U.S. government itself (intragovernmental holdings)” ×

Rewritten: “Which foreign countries hold the largest portions of U.S. federal debt, and who are the top two holders?” (by GPT-4o)

Gemini-2.5-Flash: “Japan, China” ✓

Q2 Original: “What colloquial umbrella do Rutgers and the Ivy League schools fall under?”

Gemini-2.5-Flash: “Elite schools/Top-tier universities/Ivy-level schools” ×

Rewritten: “What historical category do these schools collectively belong to, based on their founding during the colonial period?”(by GPT-4o)

Gemini-2.5-Flash: “Colonial colleges” ✓